

Make This Joystick-Controlled Robot

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The circuit presented here uses a joystick to control a robot. A joystick is more comfortable to use and accurate to control a robot as compared to switches.

The circuit is simple, compact and low-cost. It uses two ATtiny13A (eight-pin) microcontrollers (MCUs) for both transmitter and receiver circuits.

The circuit of the joystick-controlled robot comprises a transmitter and a receiver.

Transmitter. The transmitter circuit shown in Fig. 1 is built around joystick module, fixed 5V voltage regulator LM1117-5.0 (IC1), ATtiny13A MCU (IC2), 433MHz RF transmitter

module (TX1) and a few other components.

The joystick module, shown in Fig. 2, provides x-axis and y-axis control through a switch. Status of joystick movement is read by ATtiny13A MCU, which transmits the status as serial data of eight bits wirelessly through TX1.

The transmitter section gets 5V DC power supply from 9V battery through 5V regulator IC1.

Images of 433MHz RF transmitter and receiver modules are shown in Fig. 3.

Receiver. The circuit diagram of the receiver section is shown in Fig. 4. It is built around fixed 5V voltage

regulator LM1117-5.0 (IC3), ATtiny13A MCU (IC4), 433MHz RF receiver module (RX1), motor driver IC L293D (IC5), two 5V - 7.4V battery-operated geared motors (M1 and M2) and a few other components.

The receiver circuit contains another eight-pin ATtiny13A MCU (IC4). It receives eight-bit transmitted data through RX1 and then sends parallel four-bit data (D3 to D0) to IC5. It contains an H-bridge to control motors of the robot. It is powered by two power supplies, 5V and 7.4V, via jumper J1 to drive motors M1 and M2. J1 is used to select 5V motor voltage or 7.4V battery voltage.

5V regulator IC3 is used for the MCU and RX1. The 5V power supply is derived from 7.4V lithium-ion or 7.4V lithium-polymer batteries. Here, 7.4V lithium-ion battery is used.

Data transmitted from the transmitter section contains eight bits of serial data; last four bits are a replica of the first four bits. This concept is useful to cross-check the correctness of received data in the receiver section. ATtiny13A does not have serial data transmission pins; hence, a separate coding is developed for serially transmitting and receiving using ATtiny13A (like PWM concept), which is explained below.

Logic low (zero) = signal is one millisecond off with prefix one millisecond on. Logic high (one) = signal is three milliseconds off with prefix one millisecond on. Logic start (or reset) = signal is five milliseconds or more off with prefix one millisecond on. Refer Fig. 5.

Software

The source code is written in C language. Function sendCode is used in JoystickTx.C

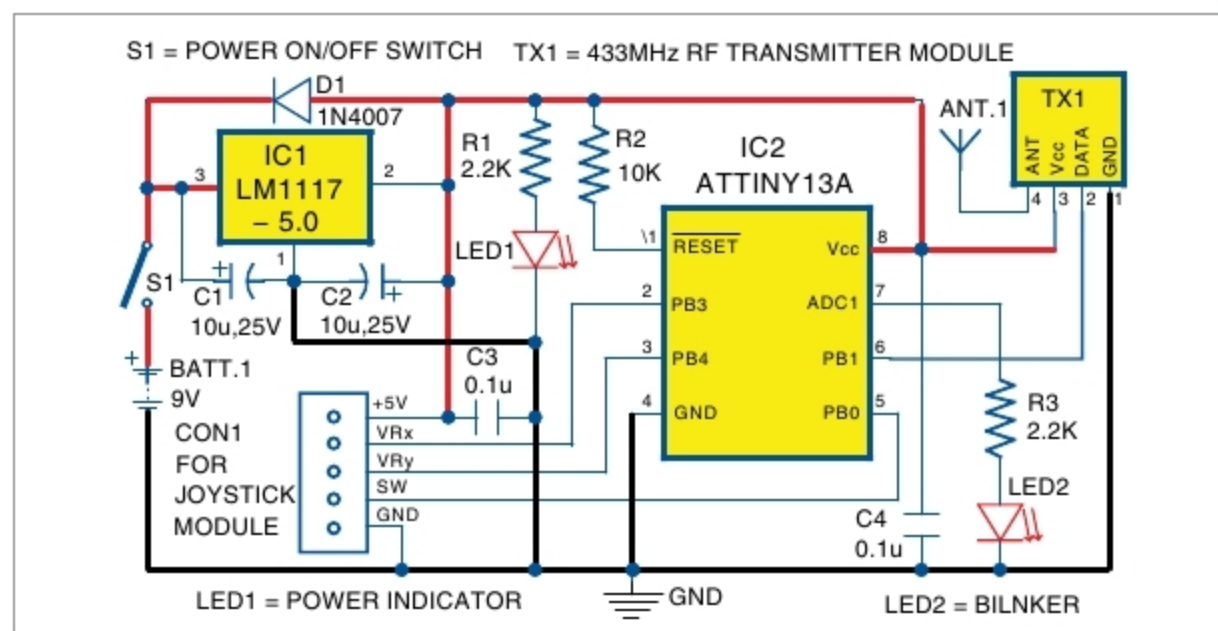


Fig. 1: Circuit diagram of the transmitter



Fig. 2: Image of the joystick

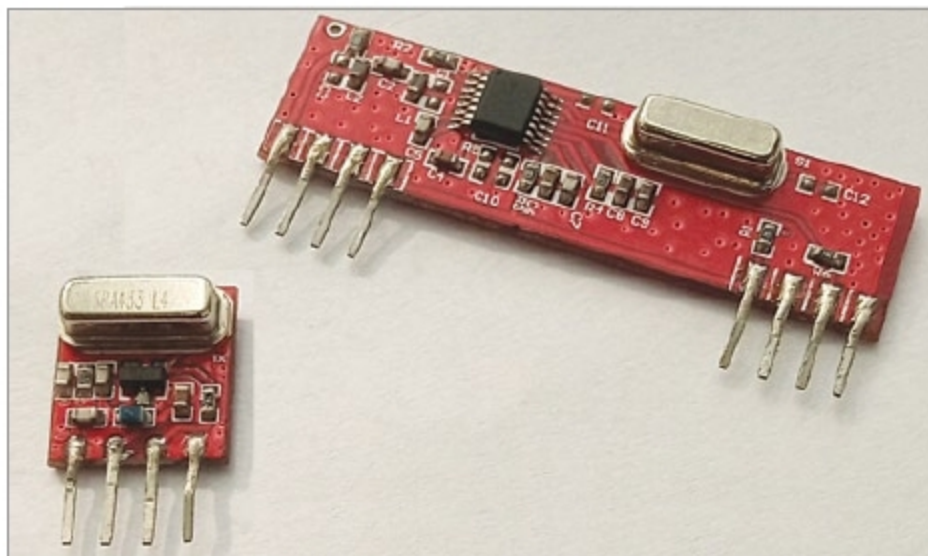


Fig. 3: 433MHz RF transmitter (left) and receiver (right) modules